

CAR-REVERSING HORN WITH FLASHER

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Here is a simple circuit that starts playing the car horn whenever your car is in reverse gear. The circuit (refer Fig. 1) employs dual timer NE556 to generate the sound. One of the timers is wired

and D2 goes high for a few seconds depending on the time period developed through resistor R4 and capacitor C4. At this point, the astable multivibrator is enabled to start oscillating. The output of the astable multivibrator is fed to the speaker through capacitor C6. The speaker,

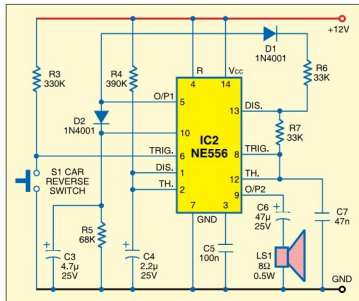


Fig. 1: Car reverse horn

as an astable multivibrator to generate the tone and the other is wired as a monostable multivibrator.

Working of the circuit is simple. When the car is in reverse gear, reverse-gear switch S1 of the car gets shorted and the monostable timer triggers to give a high output. As a result, the junction of diodes D1

and D2 goes high for a few seconds depending on the time period developed through resistor R4 and capacitor C4. At this point, the astable multivibrator is enabled to start oscillating. The output of the astable multivibrator is fed to the speaker through capacitor C6. The speaker,

in turn, produces sound until the output of the monostable is high. When the junction of diodes D1 and D2 is low, the astable multivibrator is disabled to stop oscillating. The output of the astable multivibrator is fed to the speaker through capacitor C6. The speaker, in turn, does not produce sound.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Connect the circuit to the car



reverse switch through two wires such that S1 shorts when the car gear is reversed and is open otherwise. To

power the circuit, use the car battery.

The flasher circuit (shown in Fig. 2) is built around timer NE555, which is wired as an astable multivibrator that outputs square wave at its pin 3. A 10W auto

bulb is used for flasher. The flashing rate of the bulb is decided by preset VR1.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. The flasher bulb can be mounted at the car's rear side in a reflector or a narrow painted suitable enclosure.

EFY note. A higher-wattage bulb may reduce the intensity of the headlight. You can enclose both the car-reversing horn and flasher circuits together or separately in a cabinet in your car. ●

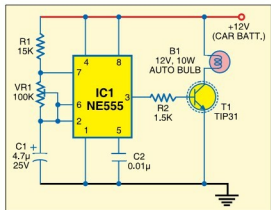


Fig. 2: Flasher circuit

SHORT-CIRCUIT PROTECTION IN DC LOW-VOLTAGE SYSTEMS

■ PRATIK PANCHAL

Many a times you need to power an adjoining accessory circuit from the power supply used in the main module circuit. Here is a circuit to derive the ad-

ditional power supply from the main circuit. The main circuit is protected from any damage due to short-circuit in the additional power supply circuit by cutting off the derived supply voltage. The derived supply voltage

restores automatically when shorting is removed. An LED is used to indicate whether short-circuit exists or not. Author's prototype of short-circuit protection module is shown in Fig. 1.

In the main power supply cir-

